Dhinesh Sadhu Subramaniam Ponnarasan

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Education

State University of New York at Binghamton | Clark Fellowship — Binghamton, NY
Master of Science - Information Systems with Applied Data Science; ***Cumulative GPA: 3.21/4.00*** — *Aug 2024 – Present*

- Coursework: Machine Learning, Generative AI, Python, JavaScript, API Development, Natural Language Processing, Deep Learning.

Vellore Institute of Technology — Bangalore, India
Post Graduate Program - Data Science; ***Cumulative GPA: 3.36/4.00*** — *Aug 2023 – June 2024*

- Coursework: Data Structures and Algorithms, Big Data, Artificial Intelligence, C/C++ Programming, Database, Data visualization

Sri Krishna Arts and Science College — Coimbatore, India
Bachelor of Computer Applications; ***Cumulative GPA: 3.20/4.00*** — *Apr 2019 – Jun 2022*

Experience

AI/ML & Applications Development Intern — Aug 2025 – Present
Uplifty AI — *Austin, Texas, United States*

- Built **AI features and applications** using **large language models (LLMs)**, implementing **retrieval augmented generation (RAG)**, **prompt orchestration**, and **tool-based workflows** with **Python** and **Azure OpenAI Service**, achieving 42% accuracy improvement through systematic **prompt design**, **prompt evaluation**, and model routing optimization.
- Designed and implemented **cloud services** on **Azure** and cloud platforms, building scalable AI applications with **Python**, **SQL**, and **Spark**, processing 500K+ records daily through distributed data processing systems, implementing **CI/CD practices**, monitoring, logging, and telemetry for production services.
- Prototyped and evaluated **AI agents** and copilot features using **function calling**, **tool use**, and **structured reasoning**, leveraging **embeddings and vector search** with **Azure AI Services**, implementing **classification**, summarization, and **NLP** workflows that achieved 88% accuracy with comprehensive evaluation pipelines.
- Implemented **Responsible AI guardrails**, security measures, and **content filtering** using **Python**, conducting **groundedness** and safety evaluations, collaborating with **cross-functional teams** including product managers to clarify requirements, communicate progress and tradeoffs, and deliver measurable outcomes using **GitHub Copilot** for accelerated development.

Software Development Engineer — June 2022 – July 2023
V3Techserv — *Chennai, India*

- Developed **cloud services** and **large-scale data processing systems** using **Python**, **SQL**, **Spark**, and cloud platforms, building predictive models and automation that improved forecasting accuracy by 34%, implementing **continuous integration and continuous delivery** pipelines with **Git version control**.
- Built end-to-end **machine learning** applications processing 2M+ records using **Python**, **SQL**, and distributed computing, implementing **data transformations**, ETL workflows with **Spark**, and creating **semantic models** for analytics, demonstrating **structured approach to system design** and service operations.
- Operated production services with **monitoring, logging, alerting**, and incident response, writing **unit tests**, **integration tests**, and implementing quality assurance practices, collaborating across teams to meet **service level objectives**, communicate technical details clearly, and deliver high-impact features in **Agile** environment.

Software Developer — Nov 2021 – May 2022
Freelancer — *Coimbatore, India*

- Built **AI applications** using **machine learning** and **Python**, implementing recommendation systems and predictive analytics that increased conversion by 28%, leveraging **SQL** databases, **data processing**, and cloud deployment with strong **analytical skills** and iterative development approach.
- Designed and implemented data-driven applications using **Python**, **SQL**, and analytics frameworks, creating automated workflows and transformations, writing tests for quality assurance, and utilizing **Git** for **version control** in collaborative development environment.
- Collaborated with stakeholders using **Agile** and **iterative development practices** to break down features, estimate work, and communicate progress, demonstrating ability to work **cross-functionally**, clarify requirements, and deliver solutions with clear documentation and technical communication.

Projects

AI Platform Services: RAG, Prompt Orchestration & Model Evaluation — Jan 2024 – Mar 2024
Python, Azure OpenAI, RAG, Prompt Engineering, Spark, SQL, Vector Search, CI/CD, Telemetry

- Designed and implemented **AI platform services** including **retrieval augmented generation (RAG)** adapters, **prompt orchestration and versioning**, and **model routing** using **Python**, **Azure OpenAI Service**, and **embeddings and vector search**, processing 1M+ records with distributed **Spark** and **SQL** achieving 89% **accuracy** and 94% **groundedness** metrics.
- Built **offline evaluation pipelines** and golden sets to measure **groundedness, quality, safety, latency, and cost**, conducting **ablation studies** and **experiment design** to compare **prompt** variants, **tool use**, and model configurations, documenting findings and graduating successful approaches into production features with comprehensive **result reporting**.
- Implemented **monitoring, logging, alerting, and tracing** for production services, writing **unit tests, integration tests**, and **prompt flow tests**, establishing **CI/CD pipelines** with **Git**, applying **Responsible AI guardrails**, security measures, and **content filtering** to ensure safe and compliant AI deployment.

AI Agents & Copilot: NLP Automation with LLMs and Tool Calling — May 2023 – Jul 2023
Python, LLMs, Azure AI, Function Calling, NLP, Spark, SQL, Prompt Design, GitHub Copilot

- Built **AI agents** and **copilot features** using **large language models**, implementing **function calling**, **tool-based workflows**, and **structured reasoning** with **Python** and **Azure AI Services**, achieving 85% F1-score through systematic **prompt design, prompt evaluation**, and multi-agent patterns exploration.
- Prototyped and evaluated emerging **AI capabilities** including **tool use**, long context strategies, and **classification and summarization** workflows, processing 500K+ documents using **Python**, **Spark**, and **SQL**, running experiments with **embeddings and vector search**, reducing processing time by 60% through optimized data transformations.
- Collaborated **cross-functionally** using **Agile practices** to clarify requirements, break down features, and estimate work, leveraging **GitHub Copilot** for code generation and tests, implementing quality measures through **integration tests** and service monitoring, communicating technical details and tradeoffs clearly to stakeholders.

Technical Skills

AI & LLM Development: Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Prompt Orchestration, Prompt Design, Prompt Evaluation, AI Agents, Copilots, Tool Use, Function Calling, Structured Reasoning, Multi-Agent Patterns
Programming & Engineering: Python, SQL, Spark, C#, JavaScript, Bash, Coding, Software Development, Technical Engineering, Data Transformations
Azure & Cloud Services: Azure OpenAI Service, Azure AI Services, Azure AI Studio, Azure Functions, Azure Machine Learning, Azure Data Explorer, Kubernetes, Microsoft Fabric OneLake, Databricks, Cloud Platforms
AI Concepts & Evaluation: Embeddings and Vector Search, Machine Learning, Groundedness Metrics, Quality Metrics, Safety Metrics, Latency Optimization, Model Routing, Prompt Flow, Evaluation Pipelines
Data Processing & Analytics: Large-Scale Data Processing, Spark, SQL, SSAS, Semantic Models, Data Transformations, ETL, Distributed Systems, Online Services
ML & NLP: Machine Learning, TensorFlow, Scikit-learn, PyTorch, Natural Language Processing, Classification, Summarization, NLP Workflows, Text Analytics, NLTK, spaCy
DevOps & Quality: CI/CD, Continuous Integration, Continuous Delivery, Git, Version Control, Unit Tests, Integration Tests, Load Tests, Monitoring, Logging, Alerting, Tracing, Service Operations
Responsible AI & Security: Responsible AI Guardrails, Content Filtering, Safety Evaluation, Risk Assessment, Red Teaming, Secrets Management, Role-Based Access, Encryption, Compliance
Development Practices: Agile, Iterative Development, Experiment Design, Ablation Studies, Research to Production, Result Reporting, GitHub Copilot, Code Generation
Collaboration & Communication: Cross-Functional Teams, Product Managers, Technical Communication, Requirements Clarification, Progress Communication, System Design, Analytical Skills
Tools & Platforms: NumPy, Pandas, Jupyter, Docker, Tableau, Power BI, PostgreSQL, Apache Spark, Service Level Objectives

Publications

Unsupervised Anomaly Detection for Network Intrusion Using Deep Autoencoders (ICNGCS 2025, IEEE)

- Built **AI application** using **machine learning** and **Python**, implementing evaluation pipelines, quality metrics, and model monitoring achieving 91% accuracy, demonstrating end-to-end development from prototype to production deployment.

Collaboration Search with Knowledge Sharing and Summarization (ICSES 2024, IEEE)

- Developed **AI features** leveraging **LLMs**, **retrieval**, and **summarization workflows**, implementing **prompt design**, **vector search**, and evaluation pipelines, collaborating cross-functionally to deliver production-ready intelligent system.

Image to Audio Conversion to Aid Visually Impaired People by CNN (ICESC 2023, IEEE)

- Implemented **machine learning application** using **Python** and cloud services, building evaluation pipelines, writing tests, and conducting experiments with structured approach to system design and quality assurance.